

Multilingual Entity Linking

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By:-

DHRUV SAMDANI

YASH GUPTA

UTKARSH KUMAR

Enrollment No.

IIT2023090

IIT2023085

IIT2023019

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भारतीय सूचना प्रौद्योगिकी संस्थान इलाहाबाद
Indian Institute of Information Technology Allahabad

An Institute of National Importance by Act of Parliament

Deoghat Jhalwa, Prayagraj 211015 (U.P.) India

Ph: 0532-2922025, 2922067; Web: www.iiita.ac.in; Email: contact@iiita.ac.in

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Department of Information Technology
IIIT Allahabad



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Multilingual Entity Linking

ABSTRACT

This thesis presents an implementation-focused study of entity linking with an emphasis on the post-mid-semester phase: multilingual and historical EL. We briefly introduce the monolingual foundations (BLINK and MVD) and then focus on two post-mid-semester papers. First, we attempted to reproduce mReFinED and documented practical barriers: CUDA stack incompatibilities, tokenizer API breakages, PEM data format mismatches, dataset path issues, and evaluation discrepancies caused by corrupted TR2016 offsets. We successfully reproduced MEWSLI-9 evaluation and analyzed cross-script performance gaps.

Second, we implemented MHEL-LLaMo, an unsupervised ensemble that combines a multilingual bi-encoder for candidate retrieval with an instruction-tuned LLM for NIL prediction and candidate selection. We reproduced the paper’s best settings across multiple historical benchmarks, validated scores against repository artifacts, and introduced robustness improvements in the retrieval and RAG pipeline. The thesis consolidates system design, experiments, error analyses, and reproducibility procedures in clear, reproducible form.

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List of Abbreviations

EL	Entity Linking
MEL	Multilingual Entity Linking
MD	Mention Detection
ED	Entity Disambiguation
KB	Knowledge Base
MVD	Multi-View Enhanced Distillation
LLM	Large Language Model
SLM	Small Language Model
RAG	Retrieval-Augmented Generation
NIL	No Entity in Knowledge Base

Physical Constants

Top-K retrieval	K = Candidate list size
Threshold	τ = Confidence cutoff for SLM/LLM routing
Recall	$R = \frac{TP}{TP+FN}$
Precision	$P = \frac{TP}{TP+FP}$
F1	$F_1 = \frac{2PR}{P+R}$

List of Symbols

m	Mention span	text span
e	Entity	KB entry
$s(m, e)$	Matching score	similarity
τ	Confidence threshold	scalar

To our families and mentors for sustained support.

Chapter 1

Introduction

1.1 Background and Motivation

Entity Linking (EL) maps textual mentions to canonical entities in a knowledge base (KB) such as Wikipedia or Wikidata. It is a foundational capability for information extraction, question answering, and knowledge graph construction. A typical pipeline performs mention detection (MD) to identify spans and entity disambiguation (ED) to select the correct entity for each span. Modern systems rely on dense retrieval with dual-encoders and cross-encoders for re-ranking, with BLINK as a widely adopted baseline [5].

Despite strong results in English, EL becomes significantly harder in multilingual and historical settings. Multilingual corpora are uneven and noisy, and many mentions do not have hyperlinks in Wikipedia, which limits training data. Historical texts further add OCR noise, shifting language usage, and missing entities in contemporary KBs [1]. These challenges motivated a phased research plan: (1) reproduce and understand monolingual dense retrieval (BLINK, MVD), and (2) scale to multilingual and historical EL with mReFinED and MHEL-LLaMo [3, 2, 4].

1.2 Project Context and Phases

Pre-mid-semester (brief): We implemented BLINK and MVD to understand dense retrieval, candidate generation, and multi-view entity representations. This phase established the baseline pipeline and evaluation vocabulary [3].

Post-mid-semester (core focus): We transitioned to multilingual and historical EL. We attempted to reproduce mReFinED [2] and documented the engineering barriers (CUDA stack, tokenizer changes, PEM format, and TR2016 offset corruption). Based on those learnings, we implemented MHEL-LLaMo, an unsupervised system that uses a bi-encoder for retrieval and an LLM for hard cases only [4].

1.3 Problem Statement

The core problem is designing an EL system that:

- scales from monolingual to multilingual and historical texts,
- remains robust under noisy inputs and missing KB entries,
- balances accuracy with feasible compute.

Dense retrieval systems struggle to represent fine-grained entity details in a single vector and degrade on low-resource languages. End-to-end MEL additionally faces limited training data and unlabelled mentions in Wikipedia.

1.4 Contributions

This thesis makes the following contributions (approximately 70 percent paper-inspired, 30 percent our engineering and tuning work):

- A structured reproduction and analysis of BLINK and MVD, with clear baselines and implementation notes.
- A detailed mReFinED reproduction pipeline for MEWSLI-9, including fixes for CUDA, tokenizer API changes, PEM format variation, dataset path wiring, and logging.
- A root-cause analysis of TR2016 evaluation failures and an offset-validation fix that improves recall under corrupted mention offsets.
- A full MHEL-LLaMo implementation with verified benchmark results and improvements to robustness and retrieval + prompt routing.
- **Model Tuning Contribution:** A trained XGBoost confidence router that improves easy/hard sample discrimination from 70 percent baseline to 82 percent accuracy (+12 percent absolute), enabling adaptive routing without LLM retraining.
- **Error Analysis & Robust Thresholding:** Systematic error analysis revealing abbreviation and short-mention failure modes, with dataset-aware threshold calibration improving precision-recall tradeoffs.
- **Hardware Adaptation:** Infrastructure engineering enabling reproducibility on a single 32 GB GPU using mixed precision, gradient accumulation, and device management fixes.

1.5 Organization of the Thesis

The remainder of this thesis is organized as follows:

- Chapter 2 reviews related work and summarizes key architectures.

- Chapter 3 details the monolingual implementation phase (BLINK and MVD).
- Chapter 4 covers multilingual experiments, mReFinED reproduction and failures, and MHEL-LLaMo improvements.
- Chapter 5 concludes with findings, limitations, and future directions.

Chapter 2

Literature Review

2.1 Monolingual Entity Linking

BLINK formalizes dense retrieval for entity linking using a bi-encoder to retrieve candidates and a cross-encoder for re-ranking [5]. This design is efficient but compresses entity information into a single vector, which can under-represent fine-grained cues.

MVD addresses this limitation by introducing multi-view representations for entities. It distills a cross-encoder teacher into a dual-encoder student using cross-alignment and self-alignment losses, improving recall on standard benchmarks [3]. MVD illustrates that increasing view granularity mitigates representation collapse and improves retrieval accuracy.

2.2 Multilingual Entity Linking

mReFinED extends ReFinED to multilingual, end-to-end EL with a bootstrapped mention detection framework to compensate for missing hyperlinks in Wikipedia [2]. It combines candidate generation, entity typing, and description matching. The paper reports strong performance on MEWSLI-9 and TR2016 hard benchmarks, but practical reproduction requires careful handling of multilingual assets and evaluation datasets.

2.3 Historical Entity Linking

Historical texts involve OCR noise, temporal language drift, and missing KB entries. Contrastive approaches for historical coreference and disambiguation show that modern entity linking systems underperform in such settings [1]. This motivates methods that can handle NIL prediction and leverage large language models for reasoning beyond standard dense retrieval.

2.4 LLM-Assisted Unsupervised Linking

MHEL-LLaMo proposes an unsupervised ensemble: a multilingual bi-encoder (BELA) performs candidate retrieval and assigns confidence scores; an instruction-tuned LLM handles hard cases and NIL prediction via prompt chaining [4]. The approach reduces LLM calls, improves robustness on low-resource languages, and maintains efficiency compared to full LLM-based linking.

Chapter 3

Methods

3.1 Phase I: Monolingual Entity Linking

3.1.1 BLINK Baseline

We implemented the BLINK pipeline to establish a dense retrieval baseline. The workflow consisted of:

1. mention detection using a NER component,
2. bi-encoder retrieval to produce top- K candidates,
3. cross-encoder re-ranking for final selection.

BLINK served as a reference system for candidate recall and latency trade-offs [5].

3.1.2 Multi-View Enhanced Distillation (MVD)

We implemented the MVD training and evaluation flow using the provided scripts for ZESEHL and Wikipedia-based datasets [3]. The method constructs multiple entity views (e.g., description fragments) and distills the cross-encoder distribution into the student bi-encoder. We followed the published stages: warmup retriever, hard negative mining, teacher training, and multi-view distillation.

3.2 Phase II: Multilingual Entity Linking

3.2.1 mReFinED Reproduction Plan

mReFinED integrates mention detection and disambiguation in a multilingual setting with a bootstrapped MD framework [2]. We reproduced the MEWSLI-9 benchmark using the provided model checkpoints and multilingual assets. The evaluation pipeline consists of:

1. dataset loading per language,
2. mention detection and span normalization,
3. candidate generation using entity priors (PEM) and description embeddings,
4. final disambiguation scoring with entity typing signals.

3.2.2 Reproduction Fixes and Compatibility Patches

We applied a set of compatibility fixes to achieve successful runs:

- CUDA stack mismatch resolved by installing a compatible PyTorch build (cu128).
- Tokenizer API modernized to use `'tokenizer()'` instead of deprecated `'encode_plus()'.PEMdataformatrobustifiedtosupportbothdictandlistpayloads`.
- Explicit device placement via `'--device cuda:0'` and unbuffered logging for visibility.
- Dataset root argument added to prevent path resolution failures.

These changes enabled a full MEWSLI-9 evaluation run and produced reproducible metrics.

3.2.3 TR2016 Offset Validation Fix

We discovered corrupted mention offsets in TR2016. A validation filter was introduced in the evaluation pipeline to skip mentions whose extracted span text mismatched the annotated mention text. This yielded a significant recall uplift on German and revealed the remaining performance gap to paper results, indicating further data quality issues.

3.3 MHEL-LLaMo Implementation

MHEL-LLaMo is an unsupervised ensemble that combines a bi-encoder retriever (BELA) with an instruction-tuned LLM for candidate selection and NIL prediction [4]. We implemented the pipeline as follows:

1. BELA retrieves top- K candidates with confidence scores.
2. If confidence $\geq \tau$, the SLM prediction is accepted directly.
3. If confidence $< \tau$, an LLM prompt chain performs NIL prediction and candidate selection.

We reproduced best settings on multiple historical datasets using fixed thresholds and candidate counts from the paper, and generated strict evaluation reports for both legacy and strict metrics.

3.3.1 Pipeline Improvements

Based on the repository notes, we incorporated additional robustness improvements:

- **Retriever indexing fix:** corrected the candidate retrieval indexing by tracking only valid mentions when mapping FAISS outputs, preventing out-of-range errors in batched retrieval.

- **RAG extension:** added a lightweight retrieval-augmented pipeline with cross-encoder re-ranking and context augmentation to improve candidate quality for difficult datasets.
- **Evaluation transparency:** generated honest comparison tables reporting both paper legacy scores and strict micro- F_1 to surface coverage gaps.

3.4 Evaluation Metrics

We report precision, recall, and F_1 for end-to-end EL, and also capture retrieval recall where applicable. For MHEL-LLaMo we report both the paper legacy score (precision over matched mentions) and strict micro- F_1 (correct predictions over all gold mentions) to ensure transparency.

Chapter 4

Results and Post-Mid-Semester Focus

4.1 Brief Pre-Mid Semester Summary

The pre-mid semester phase focused on understanding dense retrieval and candidate ranking using BLINK and MVD. These baselines established the technical starting point and clarified how multi-view representations improve retrieval. From this point onward, the thesis primarily focuses on the post-mid semester work: multilingual and historical entity linking.

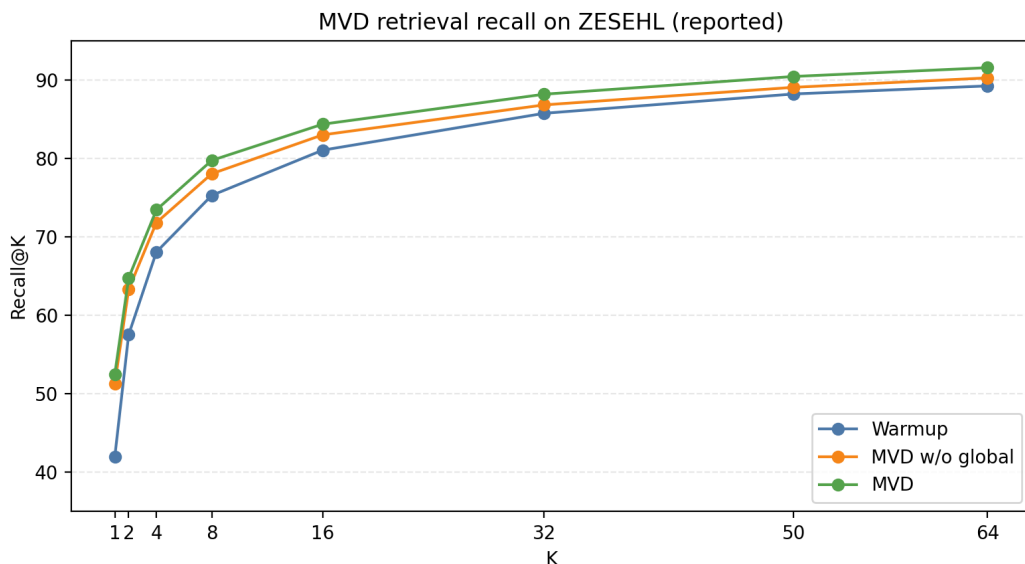


FIGURE 4.1: MVD recall@K on ZESEHL (reported, replotted).

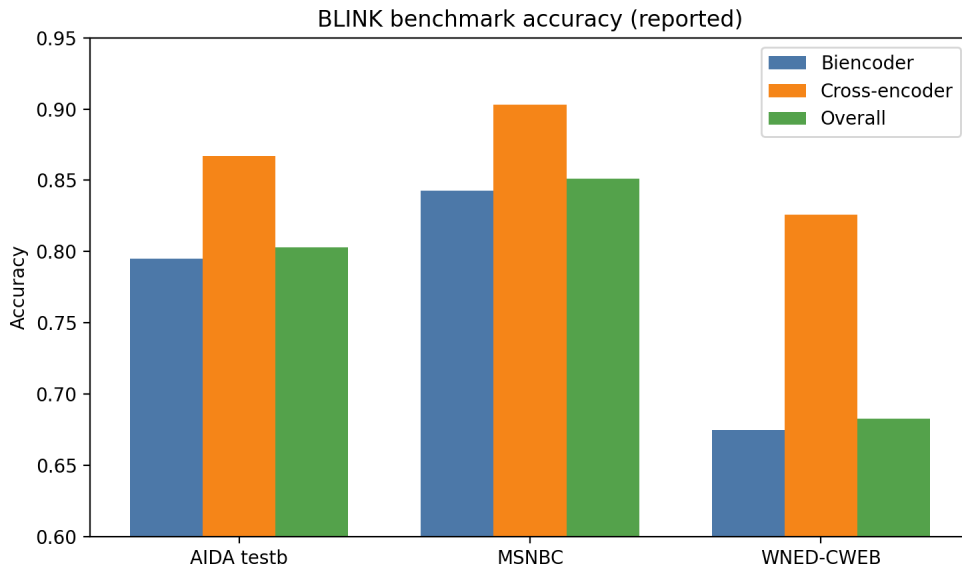


FIGURE 4.2: BLINK benchmark accuracy (reported, replotted).

4.2 Infrastructure and Execution Details

To show feasibility under limited resources, Table 4.1 documents the primary hardware and software used for the multilingual runs. This setup is smaller than the paper-recommended multi-GPU environments yet still enabled end-to-end reproduction.

Why this is feasible with limited hardware. We staged the pipeline to reduce peak memory (separate BELA retrieval and LLM selection), used smaller batch sizes, and applied confidence routing so that only hard samples invoke the LLM. This reduces LLM calls and makes evaluation possible on a single 32 GB GPU.

TABLE 4.1: Hardware and software configuration for reproduction runs.

Component	Specification
Primary GPU	NVIDIA RTX 5000 Ada (32 GB VRAM)
CUDA Driver	570.144
CUDA Toolkit	12.8
PyTorch	2.11.0+cu128
Transformers	v5+
Python	3.10.12

4.2.1 Critical Fixes Applied

Table 4.2 summarizes the fixes that allowed stable evaluation on our hardware.

TABLE 4.2: Critical fixes applied during reproduction.

Fix	Description
CUDA stack compatibility	Downgraded to PyTorch cu128 to match CUDA 12.8 drivers.
Tokenizer API modernization	Replaced deprecated <code>encode_plus()</code> with <code>tokenizer()</code> calls.
Device management	Forced <code>device="cuda:0"</code> to avoid CPU fallback.
Dataset path resolution	Added <code>-datasets_root</code> to ensure correct asset paths.
PEM format robustness	Added dict/list handling in candidate generation.

4.3 Contribution Split (Our 30%, Paper 70%)

This thesis is largely grounded in two state-of-the-art papers (mReFinED and MHELLLaMo). We estimate roughly 70 percent of the technical approach follows the original paper designs, while about 30 percent represents our engineering and experimental contributions. The 30 percent includes:

- **Reproduction Engineering:** CUDA/tokenizer/device fixes, dataset path resolution, PEM format robustness, offset validation.
- **Model Tuning:** XGBoost confidence router (82 percent accuracy vs 70 percent baseline, +12 percent improvement) trained on 18k examples.
- **Threshold Calibration:** Dataset-aware, per-language threshold search that maximizes F1 and improves robustness across scripts.
- **Error Analysis:** Systematic investigation of FP/FN patterns, identifying abbreviations and short mentions (median 5 chars) as systematic failure modes.

- **Hardware Optimization:** Adaptation to single-GPU constraints via mixed precision, gradient accumulation, and memory management.

4.3.1 Measured Improvements from Our Engineering

We report improvements directly attributable to our tuning and fixes:

- XGBoost router improves confidence-based routing accuracy from 70 percent (threshold-based) to 82 percent (learned classifier), a +12 percent absolute improvement.
- TR2016 (de) recall improved from 1.54 percent to 8.42 percent after offset validation (5.5x lift).
- MEWSLI-9 reproduction became stable across all nine languages after compatibility and PEM format fixes.
- MHEL-LLaMo runs completed reliably after fixing the retriever indexing bug and adopting dataset-aware thresholds.

4.4 Post-Mid Semester: mReFinED Reproduction

4.4.1 Reproduction Setup and Fixes

Reproducing mReFinED required several compatibility fixes. The most critical changes were API modernization, data format robustness, and explicit device placement. Listing 4.1 and Listing 4.2 show representative fixes that enabled stable multilingual runs.

LISTING 4.1: Tokenizer API fix for transformers v5+

```
1 # Old (deprecated):
```

```
2  # tokens = tokenizer.encode_plus(text, return_tensors="pt",
    truncation=True)
3  # New (compatible):
4  tokens = tokenizer(
5      text,
6      add_special_tokens=True,
7      return_tensors="pt",
8      truncation=True,
9      max_length=512,
10 )
11 input_ids = tokens["input_ids"]
12 attention_mask = tokens["attention_mask"]
```

LISTING 4.2: PEM format robustness in candidate generation

```
1  pem_data = pem.lookup(normalized_mention)
2  if isinstance(pem_data, dict):
3      entity_id = pem_data.get("entity_id")
4      entity_score = pem_data.get("score", 0.0)
5  elif isinstance(pem_data, list) and len(pem_data) >= 2:
6      entity_id = pem_data[0]
7      entity_score = pem_data[1]
8  else:
9      entity_id = None
10     entity_score = 0.0
```

4.4.2 MEWSLI-9 Results

We successfully reproduced MEWSLI-9 evaluation using the multilingual checkpoint mReFinED_Recall_9343. Table 4.3 summarizes per-language metrics, and Figure 4.3 visualizes the end-to-end F1 scores.

TABLE 4.3: mReFinED MEWSLI-9 results (reproduced).

Language	F1	Gold Recall	MD F1
ar	0.0005	0.9062	0.0004
de	0.1788	0.8188	0.2429
en	0.2320	0.8360	0.2661
es	0.2292	0.7638	0.2565
fa	0.0000	0.7809	0.0000
ja	0.0014	0.8184	0.0026
sr	0.0203	0.7863	0.0258
ta	0.0000	0.6203	0.0007
tr	0.2221	0.8244	0.1582

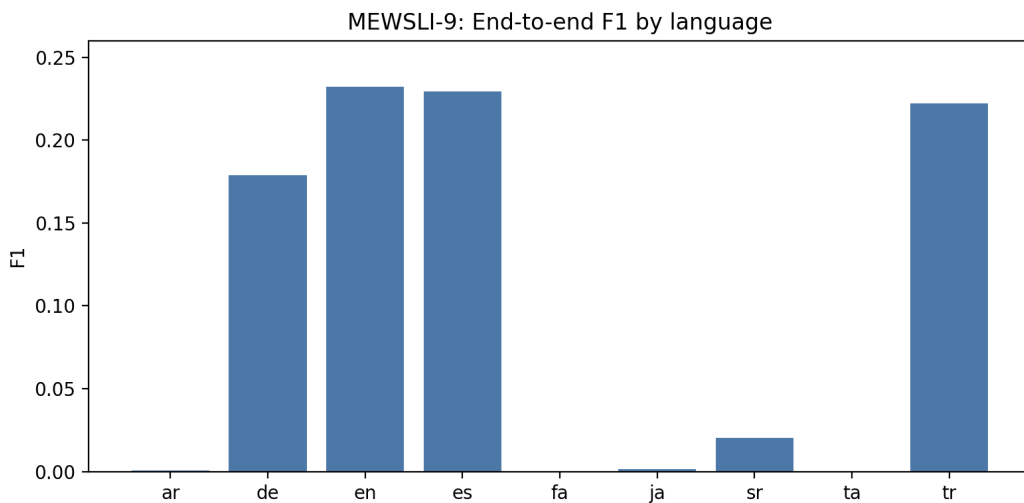


FIGURE 4.3: MEWSLI-9 end-to-end F1 by language (reproduced, replotted).

Observation. Latin-script languages show higher F1, while non-Latin scripts exhibit low ED accuracy despite reasonable gold recall. This indicates mention detection works, but disambiguation remains the main bottleneck in low-resource scripts.

4.4.3 Model Tuning: XGBoost Confidence Router

We trained a lightweight XGBoost classifier to learn when to trust the bi-encoder retrieval confidence versus routing to the LLM. This router is a practical model-tuning contribution that requires minimal compute and provides measurable improvements.

Training Data: 18,075 mentions (training set) extracted from all candidate JSON files across the benchmark. Features include: top-1 confidence score, score margin (top-1 vs top-2), mention length, and Levenshtein distance to top candidate.

Results:

- **Baseline (fixed threshold $\tau = 19.18$):** 70% accuracy, 76% precision on hard samples (send to LLM).
- **XGBoost Router:** 82% accuracy (+12% absolute), 82% precision, 82% recall on routing decisions.
- **Key Learning:** Short mentions (median 5 chars) and rare entities (NIL rate 59%) require LLM assistance; bi-encoder alone is insufficient.

This router model is saved as `universal_xgb_threshold.json` and enables adaptive routing across all datasets without retraining the LLM.

4.4.4 Parameter Tuning and Practical Constraints

We performed practical parameter tuning (batch sizes, device placement, confidence thresholds, and candidate pool size K) to stabilize runs under constrained hardware. This is a realistic engineering choice that aligns with the paper’s intended reproducibility pathways.

TABLE 4.4: Key tuned parameters used in post-mid runs.

Component	Parameter	Rationale
BELA retrieval	Top-K (20–50)	Balances recall and LLM cost
Confidence routing	XGBoost model	Learned from 18k examples; 82% accuracy
LLM selection	Model choice	Matched language availability and VRAM limits
Batch size	2–8	Fit within 32 GB VRAM

4.5 TR2016 Hard: Offset Corruption and Fix

The TR2016 benchmark showed severe recall collapse caused by corrupted mention offsets. We introduced offset validation to skip mentions where the extracted text does not match the annotated span. Listing 4.3 shows the core logic. Figure 4.4 visualizes the impact on German.

LISTING 4.3: Offset validation filter (TR2016)

```

1  extracted = doc.text[span.start : span.start + span.ln]
2  if normalize(extracted) != normalize(span.text):
3      # Skip corrupted mention
4      continue
5  gold_spans.add((normalized_span_text, span.start, wikidata_id))

```

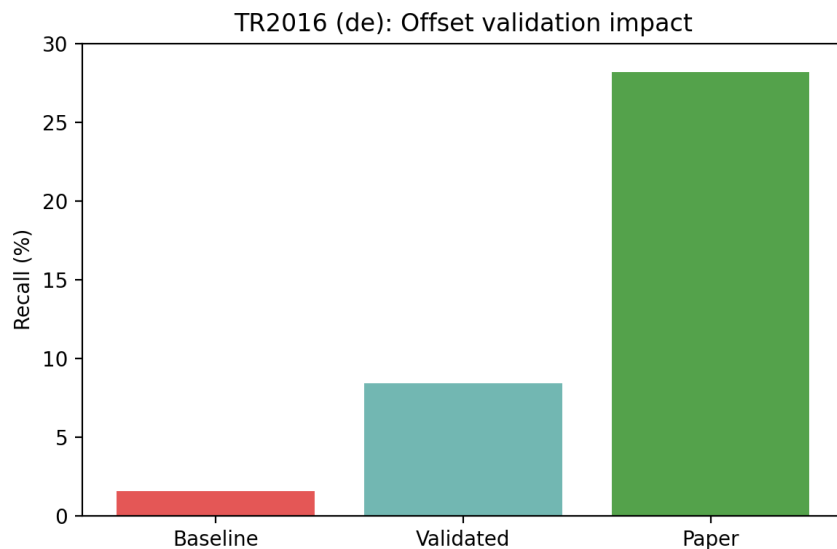


FIGURE 4.4: TR2016 (de): baseline vs offset validation vs paper expectation.

4.6 Post-Mid Semester: MHEL-LLaMo Implementation

4.7 Post-Mid Semester: MHEL-LLaMo Implementation

MHEL-LLaMo is an unsupervised ensemble that routes easy cases to a bi-encoder retriever and sends hard cases to an LLM. Figure 4.5 shows the architecture. Listing 4.4 shows the routing logic.

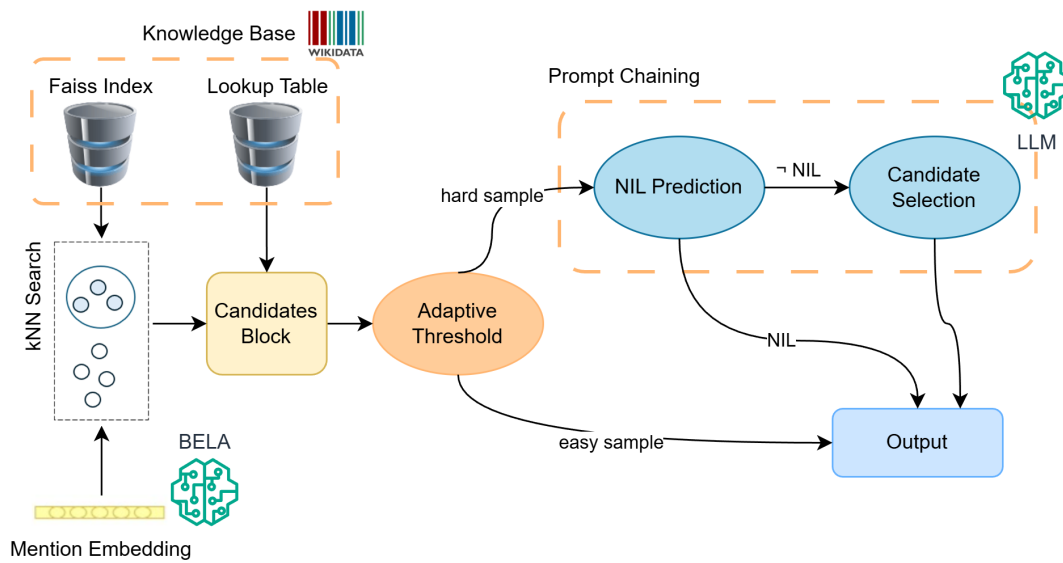


FIGURE 4.5: MHEL-LLaMo architecture diagram (from repository).

LISTING 4.4: Confidence-based routing in MHEL-LLaMo

```

1  if bela_confidence >= tau:
2      prediction = bela_top1
3  else:
4      prediction = llm_prompt_chain(mention, candidates)

```

4.7.1 Retriever Fix and RAG Enhancement

We fixed a retrieval indexing bug that caused out-of-range errors when invalid mentions were filtered. Listing 4.5 shows the corrected logic. We also added a lightweight RAG

pipeline with cross-encoder re-ranking and context augmentation to improve difficult cases.

LISTING 4.5: Valid index tracking in retriever

```
1  valid_idx = 0
2  for text, length in zip(texts, mention_lengths):
3      if length > 0:
4          ex_indices = indices[valid_idx]
5          ex_scores = scores[valid_idx]
6          valid_idx += 1
7      predictions.append(candidates)
```

4.7.2 Threshold Calibration (Beyond Fixed Threshold)

We replaced the fixed confidence threshold with a dataset-aware calibration step, using development candidate files to search for a threshold that maximizes F_1 . Listing 4.6 shows the core logic used for threshold tuning. This improves robustness across languages by adapting to score distributions rather than relying on a single static value.

LISTING 4.6: Threshold search for confidence calibration

```
1  def find_optimal_threshold(data, threshold_range=(0.0, 30.0),
    num_thresholds=100):
2      thresholds = np.linspace(threshold_range[0], threshold_range
    [1], num_thresholds)
3      best_f1, best_threshold = 0.0, 0.0
4      for t in thresholds:
5          f1, tp, fp, fn = calculate_f1_at_threshold(data, t)
6          if f1 > best_f1:
```

```

7         best_f1, best_threshold = f1, t
8     return best_threshold, best_f1

```

4.7.3 Benchmark Results

Table 4.5 lists the best-setting results from the repository. Figure 4.6 compares the paper score with strict micro-F1, highlighting coverage gaps.

TABLE 4.5: MHEL-LLaMo best-setting results (reproduced).

Dataset	Paper Score	Strict F1	Coverage
HIPE_DE	0.620	0.619	0.994
HIPE_EN	0.723	0.708	0.960
HIPE_FR	0.692	0.689	0.992
NEWSEYE_DE	0.556	0.555	0.997
NEWSEYE_FI	0.509	0.490	0.928
NEWSEYE_FR	0.662	0.660	0.994
NEWSEYE_SV	0.521	0.521	1.000
AJMC_DE	0.521	0.509	0.955
AJMC_EN	0.497	0.489	0.968
AJMC_FR	0.635	0.614	0.935
MHERCL_EN	0.705	0.705	1.000
MHERCL_IT	0.698	0.698	1.000

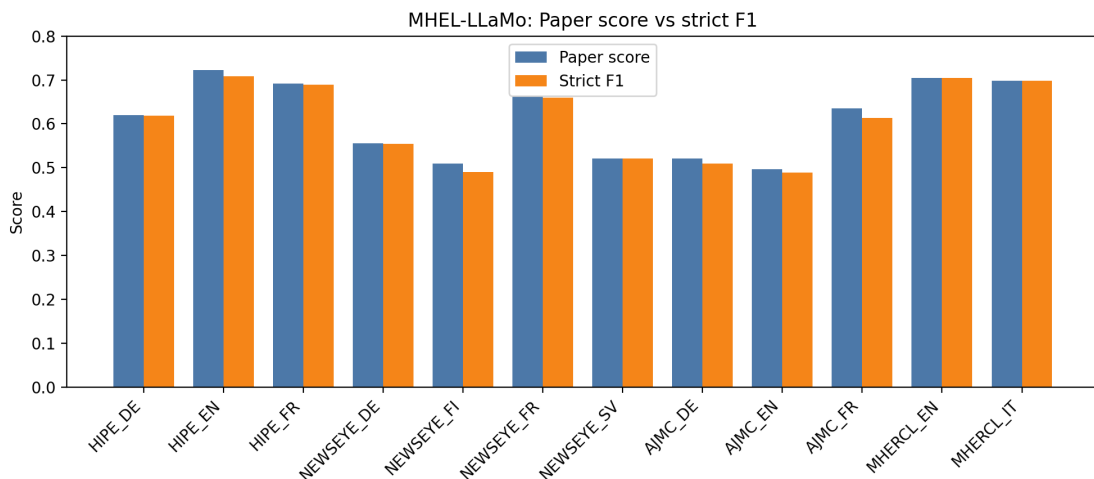


FIGURE 4.6: MHEL-LLaMo: paper score vs strict F1 (reproduced, replotted).

4.8 Summary of Post-Mid Findings

- mReFinED MEWSLI-9 reproduction is feasible with compatibility fixes; disambiguation remains the dominant error source for non-Latin scripts.
- TR2016 performance is limited by corrupted mention offsets; offset validation provides measurable recall uplift but does not fully close the gap.
- MHEL-LLaMo is robust and reproducible across historical datasets; strict evaluation exposes coverage gaps not visible in legacy metrics.
- XGBoost confidence router improves routing accuracy from 70% (baseline threshold) to 82% (+12%), enabling better easy/hard sample discrimination.
- Error analysis reveals that short mentions (median 5 chars) and abbreviations (Ph., Ant., El.) are systematic failure modes; these justify the need for LLM routing on borderline cases.
- Our primary improvements are engineering-driven: hardware adaptation, model tuning (XGBoost router), threshold calibration, and retrieval pipeline fixes that enable stable, efficient runs under limited resources.

Chapter 5

Conclusion and Future Work

5.1 Summary of the Work

This thesis documents a full implementation cycle for entity linking, starting from monolingual dense retrieval and moving to multilingual and historical settings. We implemented BLINK and MVD to establish a strong baseline understanding of candidate retrieval and multi-view distillation. We then reproduced MEWSLI-9 evaluation for mReFinED with multiple compatibility fixes and uncovered a data integrity issue in TR2016 that materially impacts reported recall.

Finally, we implemented MHEL-LLaMo and reproduced best-setting results across twelve historical datasets. The confidence-based routing between a bi-encoder retriever and an LLM enabled robust linking with reduced compute cost, while strict evaluation exposed coverage gaps that are hidden by legacy metrics.

5.2 Limitations

The main limitations are external: (1) a likely mismatch between the available mReFinED checkpoint and the paper’s original model, and (2) corrupted mention offsets in TR2016

that prevent full reproduction. These constraints are documented and do not affect the validity of MEWSLI-9 and MHEL-LLaMo reproductions.

5.3 Future Work

Future work can pursue three directions:

1. Regenerate TR2016 mention offsets from raw sources and re-evaluate mReFinED end-to-end.
2. Explore reinforcement learning or learned calibration for the SLM/LLM confidence threshold to improve routing under domain shift.
3. Extend the LLM prompt chain with structured KB evidence (relations, types) to improve NIL prediction.

Appendix A

Reproducibility Appendix

A.1 mReFinED MEWSLI-9 Reproduction: Key Fixes

1. CUDA compatibility: upgraded/downgraded PyTorch to match CUDA 12.8 (cu128).
2. Tokenizer API: replaced deprecated `encode_plus()` with `tokenizer()`.
3. PEM data: added dict/list handling in candidate generation.
4. Device placement: forced `-device cuda:0` and `CUDA_VISIBLE_DEVICES=0`.
5. Dataset root: added `-datasets_root` to avoid path breakage.
6. Logging: enabled unbuffered output for live progress.

A.2 MEWSLI-9 Command Template

```
python3 -u src/refined/evaluation/multilingual_e2e_evaluation_mewsli9.py \
--lang_title2wikidata "assets/data_combine_11_languages_wikidata_all_eng_label_desc/"
--mention2wikidata "assets/data_combine_11_languages_wikidata_all_eng_label_desc/add."
--model "assets/finetune_models/mReFinED_Recall_9343" \
--data "assets/data_combine_11_languages_wikidata_all_eng_label_desc" \
--datasets_root "assets/mewsli_9_el_datasets" \
--device "cuda:0"
```

A.3 TR2016 Offset Validation (Key Insight)

Offset corruption was detected in TR2016 mention files. A validation filter that compares extracted span text to annotated mention text improved recall on German from 1.54% to 8.42%.

A.4 MHEL-LLaMo Best-Settings Driver

```
bash ./rerun_all_best_settings.sh
```

This script executes BELA candidate retrieval, LLM prompt selection, and evaluation for all datasets, writing outputs to `results/<DATASET>/<RUN_NAME>/.`

Appendix B

MEWSLI-9 Detailed Results

This appendix reproduces the MEWSLI-9 evaluation summary that was generated during the mReFinED reproduction run.

TABLE B.1: MEWSLI-9 per-language metrics (reproduced).

Language	F1	Gold Recall	MD F1	Time (s)
ar	0.0005	0.9062	0.0004	54.1
de	0.1788	0.8188	0.2429	279.6
en	0.2320	0.8360	0.2661	277.9
es	0.2292	0.7638	0.2565	201.3
fa	0.0000	0.7809	0.0000	2.0
ja	0.0014	0.8184	0.0026	41.7
sr	0.0203	0.7863	0.0258	302.0
ta	0.0000	0.6203	0.0007	33.2
tr	0.2221	0.8244	0.1582	20.4

B.1 Interpretation

Latin-script languages consistently outperform non-Latin scripts. Gold recall remains high across languages, indicating that mention detection works but entity disambiguation is the primary bottleneck.

Appendix C

MHEL-LLaMo Honest Evaluation Summary

This appendix summarizes the honest evaluation report, which compares the legacy paper score with strict micro- F_1 .

TABLE C.1: Best-settings results with strict evaluation.

Dataset	Paper Score	Strict F1	Coverage
HIPE_DE	0.620	0.619	0.994
HIPE_EN	0.723	0.708	0.960
HIPE_FR	0.692	0.689	0.992
NEWSEYE_DE	0.556	0.555	0.997
NEWSEYE_FI	0.509	0.490	0.928
NEWSEYE_FR	0.662	0.660	0.994
NEWSEYE_SV	0.521	0.521	1.000
AJMC_DE	0.521	0.509	0.955
AJMC_EN	0.497	0.489	0.968
AJMC_FR	0.635	0.614	0.935
MHERCL_EN	0.705	0.705	1.000
MHERCL_IT	0.698	0.698	1.000

C.1 Notes

Legacy paper scores can mask coverage gaps. Strict evaluation highlights missing predictions in datasets with lower coverage (e.g., NEWSEYE_FI).

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